eedi-mining-misconceptions-in-mathematics

3rd Place Solution (with Magic Boost)

Rank	3
Team	Waseda Pochi
Author	Yi-Chia Chen
Topic ID	551498
Version	Original

Source: https://www.kaggle.com/competitions/eedi-mining-misconceptions-in-mathematics/discussion/551498
Retrieved with Kaggle CLI on 2026-10-21.

First, I want to thank the organizers for hosting this competition, and the Kaggle community for all the discussions and shared notebooks that helped me learn so much.

My Solution

A two-stage approach:

Stage 1: Retrieval

- Used Qwen-14B Embedder (from @anhvth226) ensemble with Qwen-14B embedder trained with FlagEmbedding
- Retrieved 35 most relevant misconceptions for each question
- Used publicly shared Retriever and tried my own FlagEmbedding-trained Retriever
- To be honest, my self-trained Retriever was pretty bad and removing it actually improved the score (lol)

Stage 2: Reranking

- Used Qwen-32B-instruct-AWQ reranker
- Ensembled 6 different LoRAs with various training parameters and cross-validation
- One model included about 2000 GPT4-mini generated samples, but the improvement was… not very significant

Scores at this point:

- Public LB: 0.590
- Private LB: 0.564

Time for Some Magic!

Here comes the most interesting part!

It started when I discovered over 900 unseen misconceptions in the misconceptions table that never appeared in the training data. Plus, according to @zhudong1949 and @eugenkrylov findings (check here: https://www.kaggle.com/competitions/eedi-mining-misconceptions-in-mathematics/discussion/550875), the testing set had many unseen subjects and only about 685 questions.

So I thought: these unseen misconceptions must make up a significant portion of the testing set!

Let's Experiment

On the second-to-last day of the competition (yes, I really dared to use two submissions for experiments at this point XD), I did this LB probing:

- Predicted only 1 misconception per question
- Tested twice:
    1. Using only seen misconceptions from training data: got 0.154

2. Using only unseen misconceptions: got 0.444

Seeing these results, I made a bold guess that the seen-to-unseen ratio in the testing data was roughly 1:3.

The Final Magic Touch

So I implemented a simple post-processing:

- Multiplied the probabilities of all predicted unseen misconceptions by a constant C
- Adjusted until unseen misconceptions made up 75% of the first predictions

This little magic trick made the scores skyrocket 😯 :

- Public LB: 0.590 → 0.658
- Private LB: 0.564 → 0.600

In the final version, I also randomly shuffled the order of misconceptions input to the reranker, which gave a small additional boost:

- Public LB: 0.670
- Private LB: 0.602

Code

The inference code can be found at:
https://www.kaggle.com/code/threerabbits/eedi-11-21-myq14b-q32b-rerank-mod-novel-local-suf

Supplemental Q&A

> **Ming Sheng Choo · 2024-12-14T02:38:45.657000**
>
> Gotta say this is a really interesting way to tackle the problem. Would like to ask how did you train your rerank model, was it Flagembedding or something else? Thanks!

> **Chew Kok Wah · 2024-12-14T04:14:37.537000**
>
> Do you train your Lora on AWQ weight or on the unquantized weight?

> **Yi-Chia Chen · 2024-12-14T17:21:09.313000**
>
> I train my LoRA directly on AWQ weights since I don't have enough VRAM to work with unquantized weights.

> **Yi-Chia Chen · 2024-12-14T17:23:46.253000**
>
> I used trl SFTTrainer for training, just like what many others have shared - giving numbers to 35 misconceptions (using exactly 1-9 and A-Z), and then training LLMs to output the most relevant numbers!